Threshold Experiment Report Unknown Species Detection

Myco Fungi Project

April 2, 2026

1 Objective

Detect unknown fungal species at query time using a threshold on KNN retrieval similarity scores, without any fine-tuned classifier. The system must **accept** known species (5 *Penicillium* target species) and **reject** unknown species (all other species). The optimisation target is **F1 score** on the positive (known) class.

2 Datasets

2.1 Database (Qdrant collection)

- **Collection**: `myco_fungi_features_full_finetuned`
- **Extractor**: EfficientNetB1_finetuned (1280-d)
- **Growth media**: MEA, CYA, DG18, CREA, YES, OA (all environments)

2.2 Query / Test set

- **Source**: `results/threshold/diverse_retrieval_results.csv`
- **Total samples**: 861
- **Known species** ($is\_known = 1$): 210
- **Unknown species** ($is\_known = 0$): 651
- **Retrieval**: $k = 11$, E1 (same growth medium), weighted aggregation
- **Train/test split**: One strain per species held out; all other strains of the same species remain in the database.

2.3 Known species

Penicillium citreonigrum, *P. commune*, *P. crustosum*, *P. expansum*, *P. chrysogenum*

3 Query Construction

At query time each plate image passes through:

1. **Segment** KMeans isolates the petri-dish region; 1–3 colony segments are extracted per plate.
2. **Feature extraction** Each segment is passed through EfficientNetB1_finetuned to produce a 1280-d embedding.
3. **KNN retrieval** Qdrant KNN ($k = 11$) with `env_strategy=E1` (only neighbours from the same growth medium) using named vector `EfficientNetB1_finetuned`.
4. **Sibling filtering** Same-plate neighbours are excluded.
5. **Score extraction** Top-5 neighbour cosine similarities are recorded as $s_0 \ldots s_4$ alongside each neighbour's species label.

The retrieval CSV records these five scores per query plus the ground-truth *is_known* flag (1 if the query's species is in the target set, 0 otherwise).

4 Why Autoresearch?

Manual search over formula × threshold combinations is intractable:

- The formula space is open-ended: ratio chains, gap functions, polynomial transforms, weighted sums, entropy variants, etc.
- Three threshold algorithms (`f1_grid`, `roc_opt`, `otsu`) each favour different formulas.
- There are strong interaction effects: the best formula for one algorithm is not necessarily best for another.

The autoresearch pattern addresses this with: structured logging of every (formula, algorithm, threshold) triple to `all_experiments.csv`; staircase visualisation (grey = discarded, green = new best); active reading of the log before each run to avoid repeating failures; and small focused scripts (100–800 formulas per attempt) to keep experiments interpretable.

5 Methodology Diagram

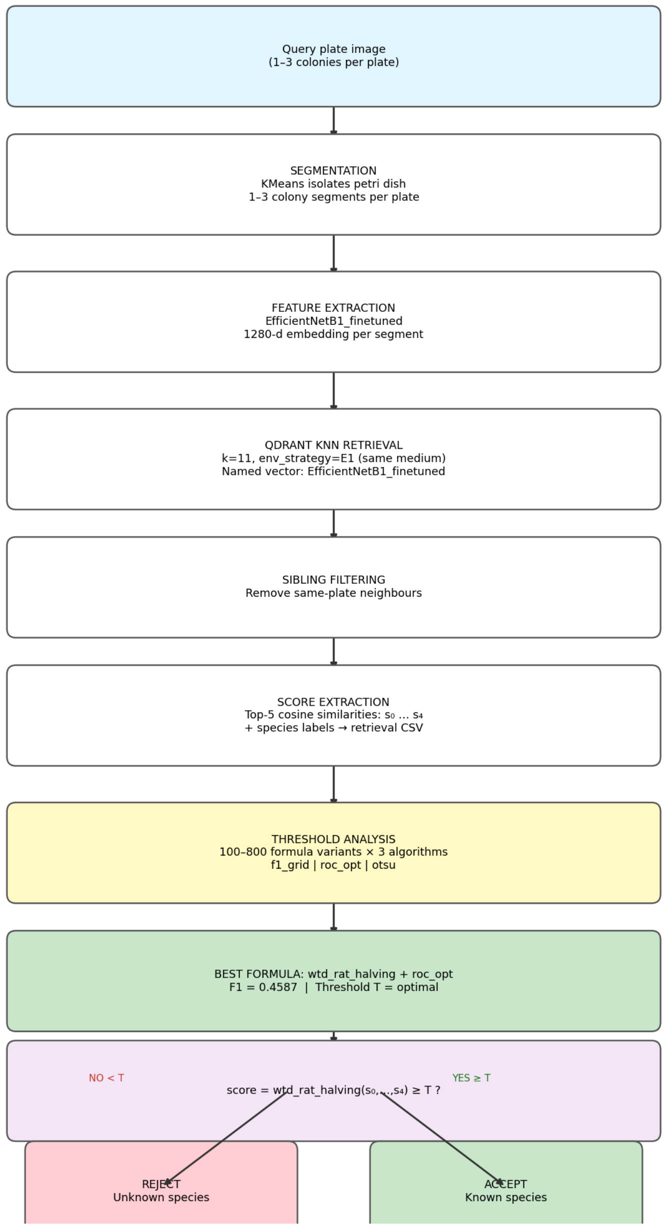


Figure 1: Retrieval pipeline (top) producing neighbour scores $s_0 \ldots s_4$, followed by threshold analysis (middle) evaluating hundreds of formula $\times$ algorithm combinations, finally producing a binary known/unknown classifier (bottom).

6 Experiment History

Att.	Date	Experiments	Best Formula	F1	Notes
1	2026-04-01	25	abs_gap_f1_grid	0.0900	baseline
2	2026-04-01	125	geom_mean_top3_roc_opt	0.1639	+0.074
3	2026-04-01	571	geom_mean_top3_roc_opt	0.1639	no change
4	2026-04-01	∼500	ratio_s0s2_roc_opt	0.4475	labels fixed
5	2026-04-01	810	rat01_p_rat12_roc_opt	0.4536	+0.006
6	2026-04-02	306	wtd_rat_halving_roc_opt	**0.4587**	+0.005

Table 1: Autoresearch history. Key turning point: Attempt 4 corrected the *is_known* label encoding, jumping F1 from ∼0.09 to ∼0.45.

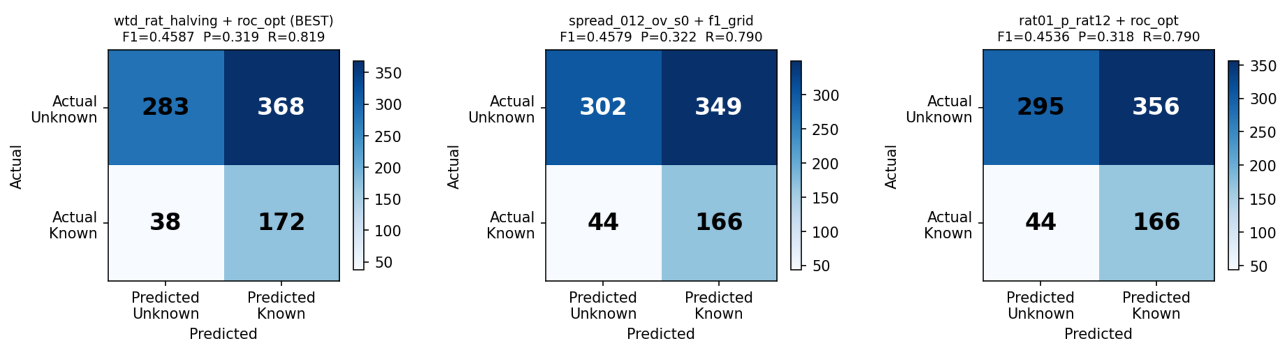
7 Best Strategies (Top-5)

Rank	Formula	Algo.	F1	P	R	Spec	TP	FP
1	wtd_rat_halving	roc_opt	**0.4587**	0.319	0.819	0.435	172	368
2	spread_012_ov_s0	f1_grid	0.4579	0.322	0.791	0.464	166	349
3	rat01_p_rat12_m_rat23	f1_grid	0.4533	0.315	0.810	0.432	170	370
4	wtd_rat_halving	f1_grid	0.4574	0.314	0.843	0.406	177	387
5	rat01_p_rat12_m_rat23	roc_opt	0.4536	0.318	0.791	0.453	166	356

Table 2: Top-5 strategies across all 2362 total experiments. The current best `wtd_rat_halving` extends the consecutive ratio idea across all 4 neighbour pairs with halving-decay weights.

8 Confusion Matrices

The three key strategies are visualised below.



Left: wtd_rat_halving + roc_opt (overall best, F1=0.4587).
Centre: spread_012_ov_s0 + f1_grid (best f1_grid, F1=0.4579).
Right: rat01_p_rat12 + roc_opt (previous best, F1=0.4536).

Figure 2: Confusion matrices for the three key strategies.

9 Formula Family Analysis

9.1 Ratio-chain family

Consecutive ratios s_i/s_{i+1} capture how much the top match dominates its immediate neighbour — a proxy for retrieval confidence.

Extended chains with decay consistently outperform the 2-term sum:

- `wtd_rat_halving` $= 1.0 \cdot (s_0/s_1) + 0.5 \cdot (s_1/s_2) + 0.25 \cdot (s_2/s_3) + 0.125 \cdot (s_3/s_4)$: F1=0.4587
- `wtd_rat_015` (base-1.5 decay, 4-term): F1=0.4530
- `rat01_p_rat12` (2-term sum, previous best): F1=0.4536
- `rat012_sum` (3-term sum): F1=0.4512

Extending to all 4 consecutive pairs with halving decay yields the best result, suggesting that diminishing contributions from distant ranks carry useful signal.

9.2 Spread / diversity family

Spread metrics (coefficient of variation of top-k scores) quantify retrieval ambiguity:

- `spread_012_ov_s0` $= \frac{\mathrm{std}(s_{0:3})}{\mathrm{mean}(s_{0:3})}/s_0$: F1=0.4579
- `gnorm02_sq` $(s_0 - s_2)^2/(s_0 + s_2)^2$: F1=0.4475

High spread indicates a clear dominant match; low spread indicates ambiguous retrieval. Normalising by s_0 removes the absolute scale, improving discrimination.

10 Key Insights

1. **Retrieval confidence is separable**: The gap/ratio between top neighbour scores is far more discriminative than raw s_0 alone. F1 for s_0 alone ≈ 0.08; best ratio-chain formulas reach F1 $\approx$ 0.46.
2. **Chain length and decay matter**: Extending consecutive ratios beyond $s_0/s_1 + s_1/s_2$ to all 4 pairs with halving-decay weights gives a small but consistent improvement.
3. `roc_opt` outperforms `f1_grid` on most top formulas by balancing sensitivity and specificity via Youden's J.
4. **Precision ceiling**: Best precision $\approx$ 32%; the main failure mode is confusing unknown species with known ones (high FP).
5. **Recall is strong**: Best recall $\approx$ 84%. The bottleneck is false positives, not false negatives.

11 Future Directions

- Train a separate binary classifier (logistic regression / SVM) on top of retrieval features rather than a single scalar threshold.
- Ensemble multiple extractors (ResNet50_finetuned + MobileNetV2_finetuned) before applying the threshold.
- Use top-1 species softmax confidence or temperature-scaled scores as additional signal.
- Investigate whether the FP unknown species share morphological similarity to the known set (e.g. shared colony texture / colour).

12 Autoresearch Staircase

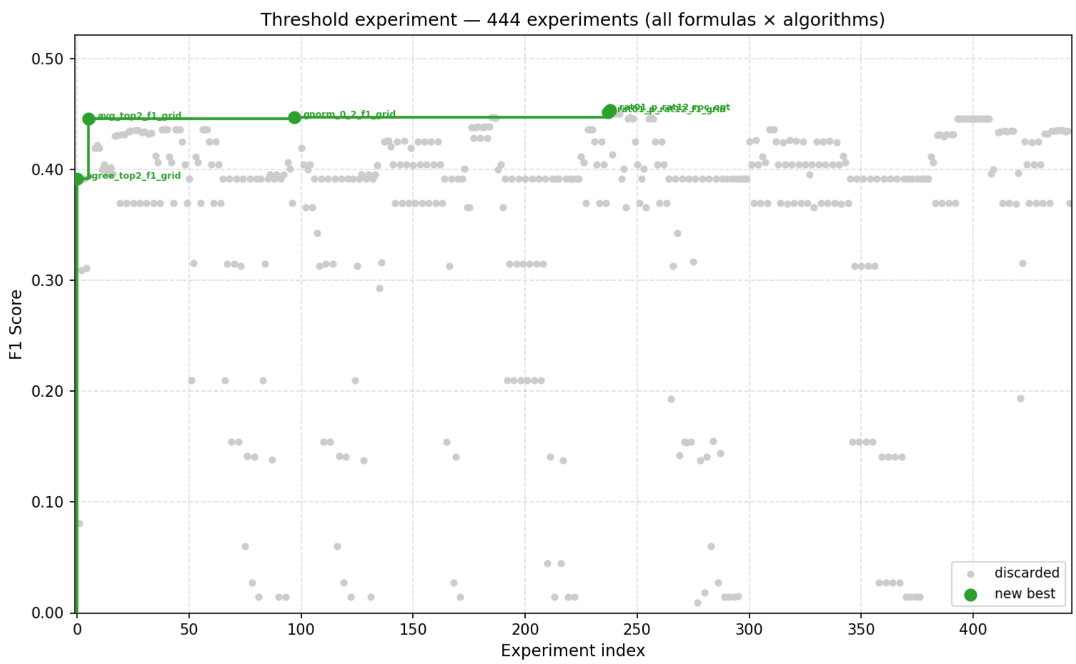


Figure 3: Autoresearch staircase chart. Each dot is one individual formula × algorithm experiment. Gray = discarded (F1 $\leq$ running best); green = new running best. The step-up pattern shows improvement over attempts.

13 Supporting Visualisations

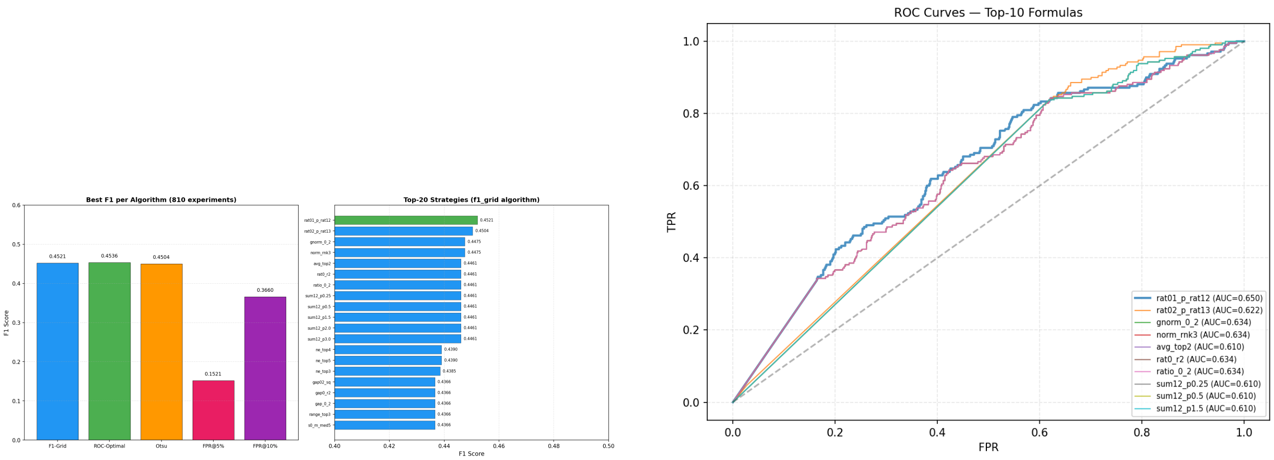


Figure 4: Left: Top-20 strategy F1 scores across all attempts. Right: ROC curves for the three best strategies.

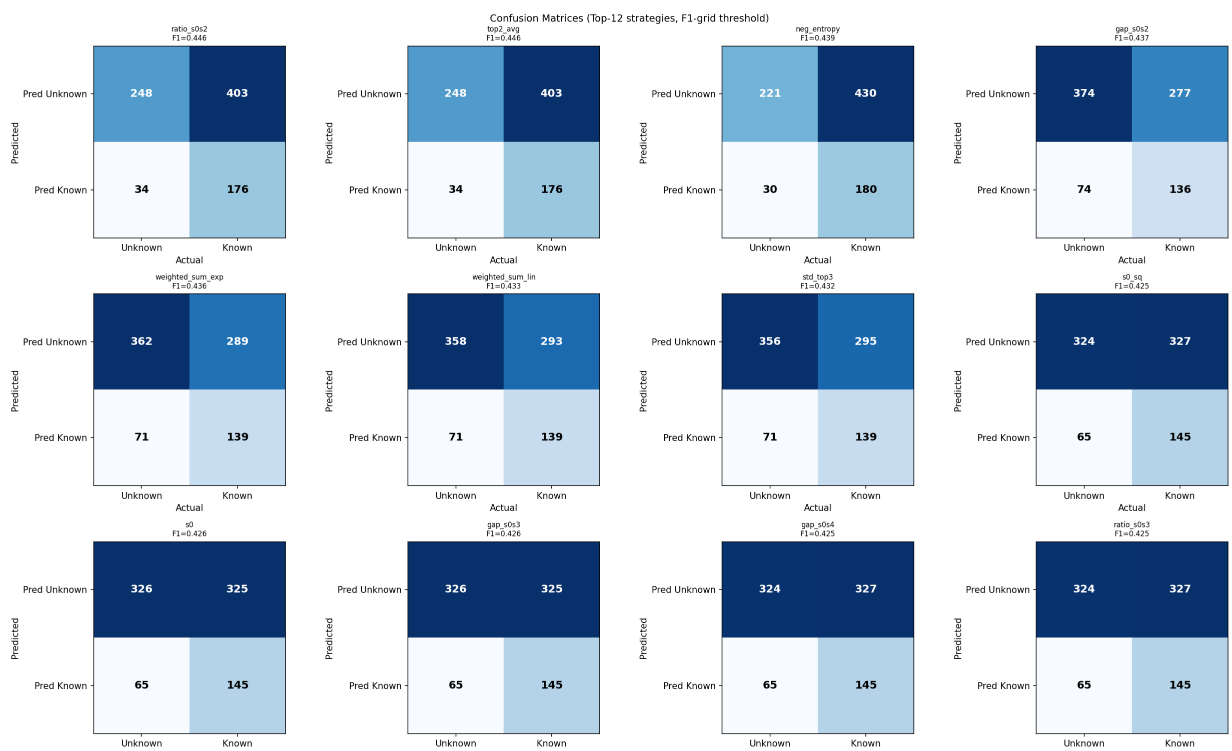


Figure 5: Confusion matrices across all top strategies in the final attempt.